Supporting Information

Mapping Losses Through Empirical Extraction of The Spatial External Luminescence Efficiency

PL setup calibration, Validation of optical constants, Regularization method, Photon recycling model and Finite elements simulation details

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S1. Photoluminsecnce system calibration

To calculate the sample ELE, a quantitative measurement of the PL photon flux must be obtained. The incident wavelength-dependent ELE measurement setup is shown in Figure 1a. The system transfer function, $T_f(\lambda)$, is the ratio between the PL spectrum, $\Phi_{out}(\lambda)$, and the spectrum recorded by the spectrometer, $\mathrm{S_{spec}}(\lambda)$. Assuming that the walls of the integrating sphere are perfectly diffusive and that PL photons are emitted isotropically from the sample into the sphere, the transfer function describing PL photons emitted from the sample is identical to that of photons reflected off a diffusive reflectance standard located at the sample port. Thus, the transfer function is:

$$T_f(\lambda) = \frac{\mathrm{S_{spec}}(\lambda)}{\Phi_{std}(\lambda) \cdot t_{int}} \qquad \text{(S1.1.)}$$

where $\Phi_{std}(\lambda)$ is the spectrum of photons reflected off the reflectance standard, and t_{int} is the spectrometer integration time. The spectrum reflected off the diffusive reflectance standard is:

$$\Phi_{std}(\lambda) = \Phi_0(\lambda) \cdot R_{std}(\lambda) \qquad \text{(S1.2.)}$$

Where $R_{std}(\lambda)$ is the reflectance standard reflectivity, and $\Phi_0(\lambda)$ is the incident photon spectrum. When illuminated by the monochromatic light source at wavelength λ_{in}, the input photon flux $\phi_0(\lambda_{in}) = \int \Phi_0(\lambda_{in},\lambda)d\lambda$ can be measured with a calibrated power meter placed at the integrating sphere sample port. Therefore, the transfer function of the system at every wavelength can be found by illuminating a diffusive reflectance standard located at the sample port with a monochromatic beam, recording the resulting spectrum, and inserting equation

(S1.2.) into (S1.1.). Since we are interested in obtaining the ELE, this process should be done at, λ_{PL}, the set of wavelengths where PL from the sample is expected:

$$T_f(\lambda_{PL}) = \frac{\int_{\lambda_{PL}-\delta\lambda}^{\lambda_{PL}+\delta\lambda} \mathrm{S_{spec}}(\lambda) d\lambda}{\phi_0(\lambda_{PL}) \cdot R_{std}(\lambda_{PL}) \cdot t_{int}} \quad \text{(S1.3.)}$$

The integral in equation (S1.3.) is needed because the input illumination is not perfectly monochromatic, and all the photons reflected off the reflectance standard should be considered. We have taken $\delta\lambda$ to be 25 nm to account for the full spectral width of the input beam. For highest accuracy, the reflectance standard should have similar reflectance to that of the sample, and the monochromator output slit should be as narrow as possible to allow differentiating between different wavelengths.

Once the system transfer function is calculated, the photon emission spectrum in response to illumination at wavelength, λ_{in}, can be found:

$$\Phi_{out}(\lambda_{in},\lambda_{PL}) = \frac{\mathrm{S_{spec}}(\lambda_{in},\lambda_{\mathrm{PL}})}{T_f(\lambda_{PL}) t_{int}} \quad \text{(S1.4.)}$$

Integrating $\Phi_{out}(\lambda_{in},\lambda_{PL})$ over all emission wavelengths results in the total emission photon flux $\phi_{\mathrm{out}}(\lambda_{in})$. The incident wavelength-dependent ELE can now be found by inserting equation (S1.4.) into equation (6). To avoid integrating sphere substitution errors, the sample, reflectance standard and power meter should be placed on different ports of the integrating sphere when each of them is measured. This assures that the total reflectance of the sphere is

identical in all three experiments (power meter, reflectance standard, and sample measurements).

S2. Optical constants validation

The sample optical properties were found using ellipsometry analysis. The ellipsometry model consisted of the GaAs substrate and a native oxide surface layer. An iterative fitting procedure was followed to find the best fit between the spectroscopic ellipsometry experimental data, the native oxide thickness, and a GenOsc model used for the substrate. The GenOsc model enables the construction of a material dielectric function by assembling different oscillator parameters [1]. The GenOsc model consists of one Drude oscillator, six PSemi oscillators, and one Gaussian oscillator. The extracted optical constants are presented in Figure S2.1. The Drude oscillator parameters used in this model are shown in Table S2.1.

Table S2.1: Drude oscillator parameters used in the ellipsometry analysis

Parameter	Parameter Input
Doping [cm^{-3}]	$1 \cdot 10^{19}$
Mobility [cm^2/Vs] ($\mu_{majority}$)	101.68 [2]
Effective mass (m)*	0.1 [2]

To verify the extracted optical constants, transmission and reflection measurements were conducted, and were compared with the results of an optical simulation. The simulated transmission and reflection spectra were calculated using GenPro4 software [3]. For this calculation, the optical constants and thicknesses of each layer were set as input. Figure S2.1b shows the simulated and measured optical transmission and reflection.

The inset in Figure S2.1a shows the absorption edge near 880 nm which is consistent with the high free carrier density of the sample [4]. The absorption of sub-bandgap photons through

free carrier absorption (FCA) results in intra-band transitions that do not lead to the generation of mobile charge carriers [5]. Hence, the spontaneous photon emission and PL spectrum are unaffected by this process. The Drude term in the optical model describes the contribution of free carriers to the dielectric function of metallic materials [6,7]. Thus, to properly calculate the PL emission spectrum, the optical constants of the sample were extracted from the same GenOsc ellipsometry model, excluding the contributions from FCA (i.e., not including the contribution of the Drude term). The difference in the optical constants can be seen in the inset in Figure S2.1a. The measured PL spectrum, and the spectrum simulated based on the modified optical constants and equation (10) are shown in Figure S2.2. The incident wavelength is 600 nm. The reflection mode configuration refers to an experiment (and simulation) in which only PL photons that are emitted from the surface on which the input illumination is incident are accounted for. In the transmission mode configuration, only photons emitted off the opposite surface are considered.

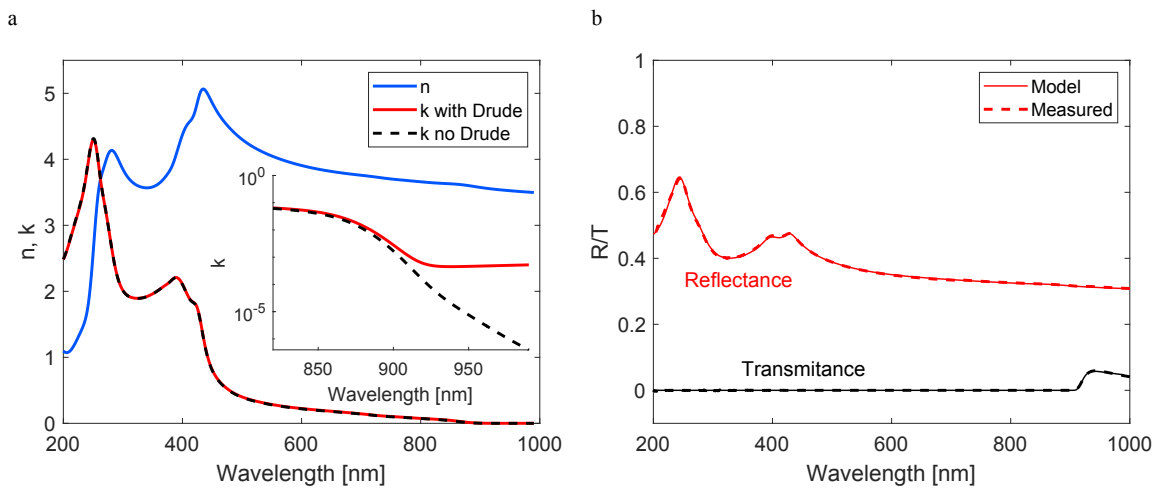

Figure S2.1. - (a) Optical constants of the GaAs wafer obtained from the ellipsometry analysis with and without the Drude oscillator which accounts for free carrier absorption. The inset

shows the extinction coefficient focusing on wavelengths near the absorption edge. (b) Modeled and measured reflectance and transmittance.

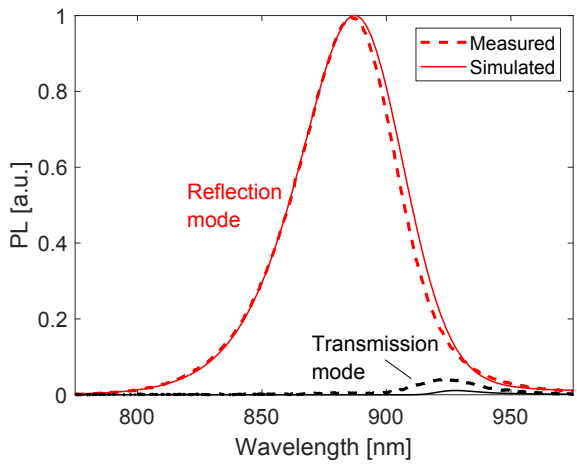

Figure S2.2. - Simulated and measured PL spectra in reflection (when collecting photons emitted from the surface on which the input illumination is incident) and transmission mode (when collecting photons emitted from the surface opposite to the surface on which the input illumination is incident). Both measured spectra were normalized by the maximum value of the measured reflection-mode PL spectrum, and both simulated spectra were normalized by the maximum value of the simulated reflection-mode PL spectrum. The incident wavelength is 600 nm.

S3. The regularization method

A regularization process was performed to extract the SELE profile, as detailed in the methods section. In the regularization process, the candidate SELE profiles are obtained by minimizing the problem shown in equation (8) for different values of the weight variable, κ. Small values of κ produce solutions that accurately reconstruct the measurement but may result in an unphysical solution. Large values of κ result in over-constrained solutions. These solutions might have a physically acceptable shape and magnitude but cannot reconstruct the measured data with an acceptable accuracy. The desired solution is a compromise between solutions that are constrained to have physical features, and solutions that reconstruct the measurements accurately.

For every value of κ, the reconstruction error $\epsilon = \left\| \frac{\overline{\overline{\Delta G}}(\lambda_{in}, z_i)\bar{S}(z)}{\bar{\phi}_{abs}(\lambda_{in})} - \bar{\eta}_{ext}(\lambda_{in}) \right\|_2$, and the constrained candidate solution, $\left\| L\left(\bar{S}(z)\right) \right\|_2$, are calculated. The L-curve is a plot of ϵ as function of $\left\| L\left(\bar{S}(z)\right) \right\|_2$. The knee of the curve corresponds to κ values that lead to solutions that both have physical features and can reproduce the measurements with an acceptable accuracy. κ_{knee} is the weight variable that corresponds to the knee of the L-curve. Figure S3.1. shows the L-curve obtained in the regularization process. The chosen profile is the mean of the solutions with κ values in the range of $\sqrt{10}\kappa_{knee} < \kappa < \kappa_{knee}/\sqrt{10}$.

Because the sample thickness is considerably larger than the absorption depth for all incident wavelengths, there is no optical generation in significant portions within the device. Hence, the regularization process cannot provide information on the SELE in these regions. As a result, constraining the magnitude of the solutions leads to an unphysically low SELE in these un-

interrogated regions. PL measurements in transmission mode (illuminating one surface of the sample and measuring PL off the other) show a non-negligible PL signal. Thus, the SELE should not be zero deep within the device bulk (as seen in Figure S2.2.). Furthermore, since there is negligible optical generation deep within the device, the optical generation matrix $\overline{\overline{\Delta G}}$ is practically zero in this region leading to high uncertainty in the reconstructed $\overline{S}(z)$ deep within the sample. This issue is insignificant for practical solar cells where the optical generation is non-zero at every depth within the device.

The choice of mesh affects the extracted SELE profiles and must be designed with great attention. Near the surface, where most of charge generation occurs, the mesh is uniform. Deeper within the wafer, where the optical generation is significantly lower, the size of the mesh elements follows a logarithmically spaced array. The logarithmically spaced sub-mesh is chosen so that its first element is the size of the linear sub-mesh elements. The length of the linear sub-mesh is chosen according to maximum absorption depth for the highest incident wavelength, and the number of elements in the linearly spaced mesh is chosen to produce the lowest RMSE between the ELE measurement and the reconstruction according to equation (2). The number of elements in the second mesh, was chosen as the smallest number that results in a non-negative SELE throughout the material. Figure S3.2. shows the magnitude of the mesh elements.

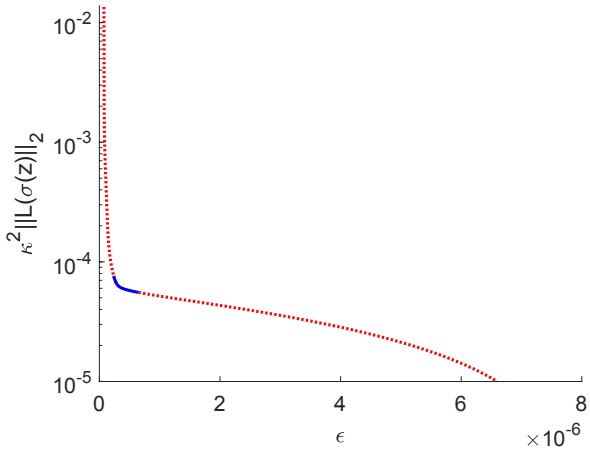

Figure S3.1. – The regularization L-curve diagram. The thicker line marks the region with $\sqrt{10}$ $\kappa_{knee} < \kappa < \kappa_{knee}/\sqrt{10}$.

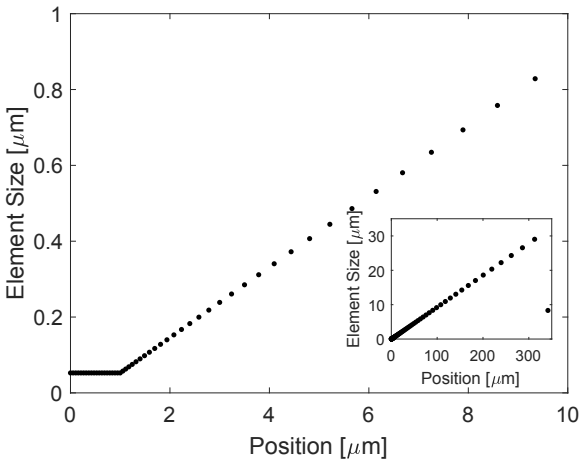

Figure S3.2. – Mesh elements size and position. The elements near the surface are uniform in size. The rest of the elements up to the sample thickness are logarithmically spaced.

S4. The photon recycling model

When charge carriers recombine radiatively, photons that are generated within the material propagate until they are reabsorbed or until reaching an interface off which they are reflected or transmitted. Photons that are transmitted to the ambient are those that contribute to the PL spectrum and to the ELE. However, to accurately simulate the emitted PL flux, photon reabsorption within the material and the resulting optical generation processes (photon recycling) should be accounted for. Photon recycling effects were simulated by coupling a ray optics simulation to a finite elements simulation which models charge transport throughout the device (section S5.). The coupling of the two simulations was done with an iterative algorithm similar to the one reported by Walker et. al. [8].

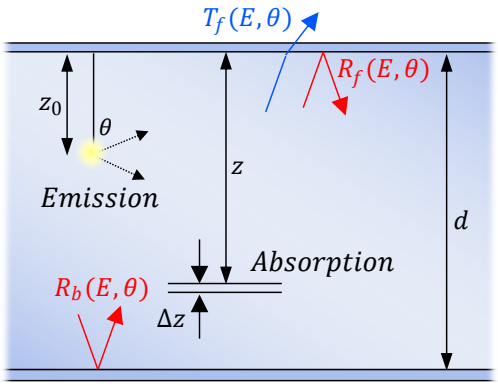

Figure S4.1. – Photon emission and absorption schematics in a 2D infinitely wide device of thickness d. A photon is generated at a distance z_0 from the front surface and absorbed in a small depth Δz at a distance z. $R_f(E,\theta)$, $R_b(E,\theta)$ and $T_f(E,\theta)$ are the front and back reflectance and transmission Fresnel coefficients, respectively.

The optical model used for the photon recycling simulation assumes a flat and homogenous material of thickness d, with a thin native oxide layer on each surface, with no absorption in the oxide layers. Figure S4.1. shows a schematic diagram of the processes that photons undergo

within the material. Assuming Beer-Lambert optics, and that the generated photons propagate isotropically, the fraction of photons that are absorbed and contribute to optical generation in a small element with thickness Δz, located at a distance z from the front surface, prior to any reflections, are:

$$A^{dir}(\theta,z,z_0,E) = \frac{\alpha_b}{\alpha}\begin{cases} e^{-\frac{\alpha(E)(z_0-(z+\Delta z))}{cos(\theta)}} - e^{-\frac{\alpha(E)(z_0-z)}{cos(\theta)}} & cos(\theta) > 0 \\ e^{-\frac{\alpha(E)(z-z_0)}{|\,cos(\theta)\,|}} - e^{-\frac{\alpha(E)((z+\Delta z)-z_0)}{|cos(\theta)|}} & cos(\theta) < 0 \end{cases} \quad \text{(S4.1.)}$$

θ is the angle between the photon propagation direction and the normal to the front surface, and z_0 is the depth in which the photon was generated. The pre-factor α_b/α is used to account for photon absorption events that contribute to the generation of mobile charge carriers and exclude free carrier absorption (see section S2.). The fraction of photons that are reflected once prior to being absorbed in Δz is:

$$A_1^{ref}(\theta,z,z_0,E) = \frac{\alpha_b}{\alpha}\begin{cases} \left(e^{-\frac{\alpha(E)(z_0+z)}{cos(\theta)}} - e^{-\frac{\alpha(E)(z_0+(z+\Delta z))}{\cos(\theta)}}\right)R_f(\theta,E) & \cos(\theta) > 0 \\ \left(e^{-\frac{\alpha(E)(2d-z_0-(z+\Delta z))}{|cos(\theta)|}} - e^{-\frac{\alpha(E)(2d-z_0-z)}{|\cos(\theta)|}}\right)R_b(\theta,E) & \cos(\theta) < 0 \end{cases} \quad \text{(S4.2.)}$$

$R_f(\theta,E)$ and $R_b(\theta,E)$ are the angle dependent reflectivity of the front and back surfaces respectively for the reflectance of a photon with energy E. $R_f(\theta,E)$ and $R_b(\theta,E)$ were calculated using the transfer matrix method (TMM) [9,10]. The absorbance for photons undergoing two reflection events is:

$$A_2^{ref}(\theta,z,z_0,E) = \frac{\alpha_b}{\alpha}\begin{cases} \left(e^{-\frac{\alpha(E)(2d+z_0-(z+\Delta z))}{cos(\theta)}} - e^{-\frac{\alpha(E)(2d+z_0-z)}{\cos(\theta)}}\right)R_f(\theta,E)R_b(\theta,E) & \cos(\theta) > 0 \\ \left(e^{-\frac{\alpha(E)(2d-z_0+z)}{|cos(\theta)|}} - e^{-\frac{\alpha(E)(2d-z_0+(z+\Delta z))}{|\cos(\theta)|}}\right)R_b(\theta,E)R_f(\theta,E) & \cos(\theta) < 0 \end{cases} \quad \text{(S4.3.)}$$

This analysis is repeated for higher orders of reflection, to account for the contribution of photons with energies that are near the band gap energy and are hardly absorbed. At $\theta = \pm 90°$ $A^{dir} = \alpha_b/\alpha$ for $z = z_0$, and $A^{dir} = 0$ for all other values of z. Since at $\theta = \pm 90°$ photons are not reflected off any surface, $A_i^{ref}(\pm 90°, z_0, z, E) = 0$. Thus, the probability for a photon generated at z_0 to be reabsorbed in an element Δz at depth z from the front surface, $P_{PR}(z, z_0, E)$, is the absorptance at z accounting for all reflection orders and propagation angles.

$$P_{PR}(z,z_0,E) = \frac{1}{2\pi}\int_0^{2\pi} A^{dir} + \sum_i A_i^{ref}\ d\theta \qquad \text{(S4.4.)}$$

The normalization by 2π is required to account for the isotropic emission of radiative recombination photons. The optical generation due to photons that are reabsorbed at depth z, $G_{PR}^i(z)$, then follows:

$$G_{PR}^i(z) = \int_0^{\infty}\int_0^{d} \frac{1}{\Delta z} dR_{rad}^i(E,z_0) P_{PR}(z,z_0,E) dz_0 dE \qquad \text{(S4.5.)}$$

Where $dR_{rad}^i(E,z)$ is the photon generation rate calculated according to equation (9). The superscript i is the photon recycling order as explained below. The total optical generation rate is the generation rate induced by the incident illumination, $G_0(z)$, with the contribution from photon recycling:

$$G_{tot}^i(z) = G_0(z) + G_{PR}^i(z) \qquad \text{(S4.6.)}$$

The optical generation due to photon recycling augments the electron-hole distribution, which in turn determines the distribution of photons generated through radiative recombination. Thus, the continuity and Poisson equations must be coupled to the equations

describing photon recycling. This coupling was implemented with an iterative process as illustrated in Figure S4.2.[8] The Poisson and continuity equations are solved using a COMSOL Multiphysics simulation. From this simulation, the depth dependent quasi-Fermi levels separation is computed and inserted into equation (9), which together with $P_{PR}(z,z_0,E)$, is used to compute an updated value for $dR^i_{rad}(E,z_0)$ and $G^i_{PR}(z)$. The updated optical generation profile $G^i_{tot}(z)$ is inserted back into the COMSOL simulation and the charge carrier distribution is recalculated. This process is repeated until convergence.

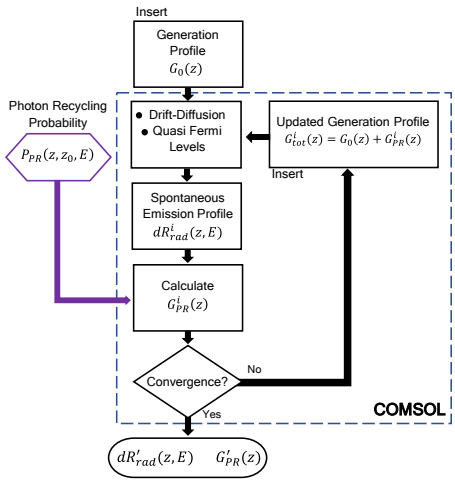

Figure S4.2. - A flow chart of the iterative process for coupling photon recycling with the charge transport simulations.

Once the solutions converge, the PL spectrum and the ELE can be calculated according to equation (10). The total photon escape and reabsorption probabilities follow:

$$P_{abs}(z,E) = \int_0^d \frac{1}{\Delta z} P_{PR}(z,z_0,E)\, dz_0 \qquad \text{(S4.7.)}$$

$$P_{esc}(z,E) = 1 - \int_0^d \frac{1}{\Delta z} P_{PR}(z,z_0,E)\, dz_0 \qquad \text{(S4.8.)}$$

Since PL measurements are conducted in reflection or transmission mode, $P_{esc}(z,E)$ should be divided to the escape probability from the front and back surface ($P_{esc}^{front}(z,E)$ and $P_{esc}^{back}(z,E)$ respectively) following [11]:

$$P_{esc}^{front}(z,E) = \int_0^{2\pi}\int_0^{\pi} T_f(\theta,E) \frac{\left(e^{-\frac{\alpha(E)z}{\cos(\theta)}} + R_b(\theta,E)e^{-\frac{\alpha(E)(2d-z)}{\cos(\theta)}}\right)}{1 - R_f(\theta,E)R_b(\theta,E)e^{-\frac{\alpha(E)2d}{\cos(\theta)}}} \sin(\theta)d\theta d\varphi \qquad \text{(S4.9.)}$$

$$P_{esc}^{back}(z,E) = \int_0^{2\pi}\int_0^{\pi} T_b(\theta,E) \frac{\left(e^{-\frac{\alpha(E)(d-z)}{\cos(\theta)}} + R_f(\theta,E)e^{-\frac{\alpha(E)(d+z)}{\cos(\theta)}}\right)}{1 - R_b(\theta,E)R_f(\theta,E)e^{-\frac{\alpha(E)2d}{\cos(\theta)}}} \sin(\theta)d\theta d\varphi \qquad \text{(S4.10.)}$$

$T_f(\theta,E)$ and $T_b(\theta,E)$ are respectively the front and back surface angle-dependent transmittance for a photon with energy E, calculated the same way as $R_f(\theta,E)$ and $R_b(\theta,E)$. Figure S4.3. shows a comparison between the wavelength dependent photon escape and absorption probabilities, averaged over the thickness of the sample, and the escape and absorption probabilities, calculated using the model presented by Steiner et. Al. [11]. Free carrier absorption was not considered in this comparison. This agreement between the two calculations verifies the validity of the presented model.

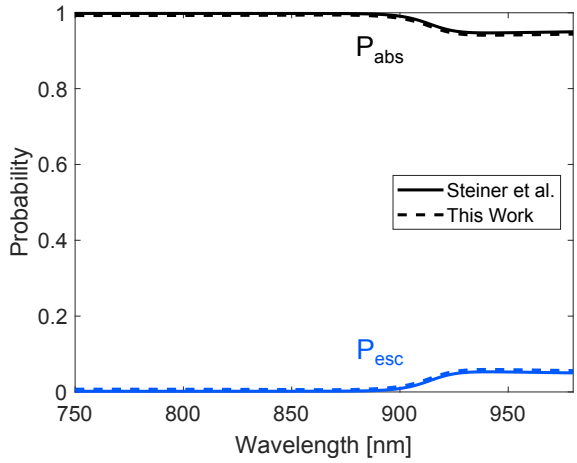

Figure S4.3. – Photon absorption and escape probabilities calculations, $P_{abs}(E)$, $P_{esc}(E)$ using the photon recycling model (presented in this work) and following Steiner et al. [11]. The probabilities are averaged over the sample thickness.

Figure S4.4a shows the band diagram of a p-type GaAs wafer when photon recycling effects are considered and when they are not. Photon recycling dramatically increases the quasi-Fermi level separation deep within the material. Figure S4.4b shows the simulated reflection and transmission (inset) PL spectrum with and without the iterative photon recycling calculation. As expected, the introduction of photon recycling effects increases the PL intensity. However, since the escape probability deep within the material is small, the large quasi-Fermi level separation induced by photon recycling does not contribute as much to the PL intensity in reflection mode.

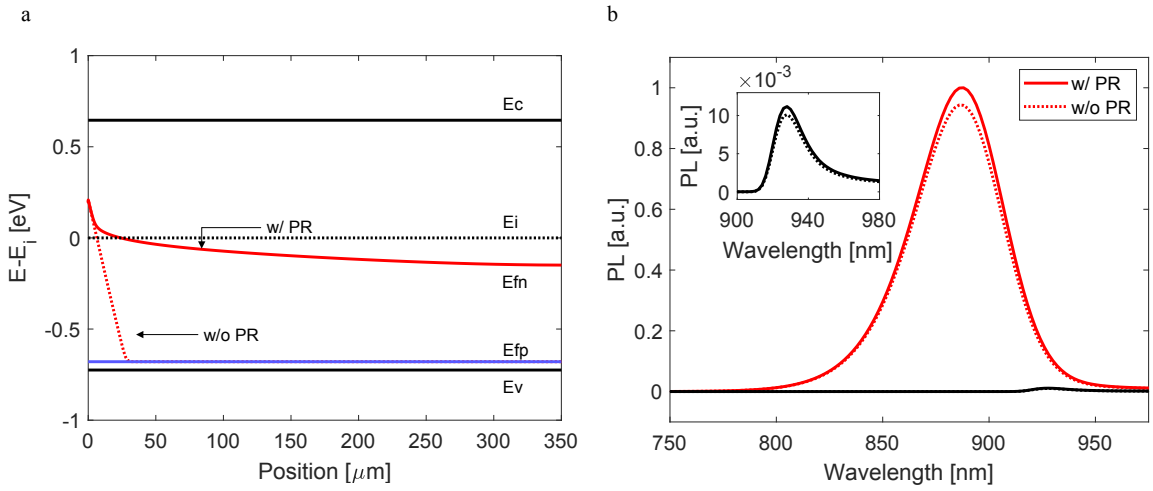

Figure S4.4. – (a) Energy band diagram with and without the inclusion of photon recycling effects. (b) Simulated PL spectrum (normalized) with and without the inclusion of photon recycling effects. Inset shows the PL spectrum in transmission mode. The sample properties are detailed in Table S5.1. and the incident wavelength is 600 nm.

S5. The finite elements simulations

The coupled Poisson and continuity equations for electrons and holes were solved numerically using a commercial finite elements partial differential equations solver software (COMSOL Multiphysics V6.1). The simulation domain is a one-dimensional, uniform cross-section of a GaAs wafer with a thickness of 350 μm. The potential boundary conditions applied a reference potential of 0V at the surface facing the illumination and zero field at the other surface. Surface recombination was used for the boundary conditions for electrons and holes at both surfaces. The material properties and the optical generation profile produced by the incident monochromatic illumination, $G(z)$, are inputs to the simulation. To account for degeneracy effects, Fermi-Dirac statistics are assumed. Radiative, Auger, and SRH recombination were assumed for the bulk material.

Due to higher excess of carriers near the surface, the mesh elements length was chosen to follow a logarithmic scale, expanding from the surface of the device to its end. The same mesh was used in all calculations except for the regularization process (see section S3.). Table S5.1. lists all the material properties used in the simulation. The material properties were initially taken from the referenced values and those were modified to find the best fit between ELE measurements and simulation. The carrier concentration and thickness of the device are determined based on the sample properties given by the manufacturer. Deviations from measured and expected values might be due to inaccuracies in the optical absorption coefficients obtained from the ellipsometry measurements. Because of the sensitivity of the PL

to the absorption coefficient at energies near the band gap, any deviation in absorption coefficients results in a relatively large deviation in other material properties.

Table S5.1.: GaAs material properties used in simulation

Parameter	Best fit	Referenced Value {sample spec.}
Doping [cm^{-3}]	$1 \cdot 10^{19}$	{(1.74±0.83) ·10^{19}}
Thickness [μm]	350	{364±2}
Mobility [cm^2/Vs] (majority, minority)	(101.68 [2], 1500)	({71±10}, 1000 [12–15])
Radiative recombination coefficient [cm^6/s]	$2.0 \cdot 10^{-10}$	$1 \cdot 10^{-10}$ [16]
Auger recombination coefficient [cm^6/s]	$15 \cdot 10^{-30}$	$10 \cdot 10^{-30}$ [17,18]
SRH lifetime [ns] (holes, electrons)	(0.4,0.4)	(0.82, 1) [19]
Surface Recombination velocity [cm/s]	$2 \cdot 10^{7}$	$1 \cdot 10^{7}$ [20,21]
Band gap narrowing model	Jain-Roulston[22]	-

The SELE of the device was then calculated using equation (5), where an additional optical generation at point z, $\Delta G \cdot \delta(z - z')$ was added to the optical generation profile, induced by the incident illumination. To assure that the delta-function generation is only a small signal and does not have a significant effect on the energetics of the device, the size of ΔG is set to be 10% of the total optical generation in the sample:

$$\int_0^d \Delta G \cdot \delta(z - z')dz = 0.1 \cdot \int_0^d G(z)dz \qquad \text{(S5.1.)}$$

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